Supplementary Materials to "Covariate-Assisted Bayesian Graph Learning for Heterogeneous Data"

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S.1 Proof of Lemma 3.1

Proof. For any $\epsilon > 0$, let $K_\epsilon = \left(\frac{q^2 H_c (b-a)^{\nu+2}}{\epsilon^2}\right)^{1/\nu}$, where $H_c = \max_{i,j}\{\|\sigma_{ij}^0\|_{\mathcal{H}^\nu}\}$. Choose $a_k = a + \frac{b-a}{K_\epsilon}(k-1)$, then $a_0 = a$ and $a_{K_\epsilon+1} = b$. Let $\Delta_k = (a_k, a_{k+1}]$.

Given any $i, j \in \{1, \ldots, q\}$, for $\sigma_{ij}^0(x)$, we first choose a piecewise linear function $\sigma_{ij}(x) = \sum_{k=1}^{K_\epsilon} \delta_{ij}^k \mathbb{1}_{\Delta_k}(x)$ such that

$$|\sigma_{ij}^0(x) - \sigma_{ij}(x)| \le \|\sigma_{ij}^0\|_{\mathcal{H}^\nu} \frac{b-a}{K_\epsilon}, \quad \text{for any } x \in (a, b].$$

To show this, set $\delta_{ij}^k = \sigma_{ij}^0(a_k)$. From the ν-Hölder continuity,

$$|\sigma_{ij}^0(x) - \sigma_{ij}^0(x')| \le \|\sigma_{ij}^0\|_{\mathcal{H}^\nu} \cdot |x - x'|^\nu, \quad \text{for any } x, x' \in (a, b].$$

Then, for $x \in \Delta_k$, $k = 1, \ldots, K_\epsilon$, we have

$$|\sigma_{ij}^0(x) - \sigma_{ij}(x)| = |\sigma_{ij}^0(x) - \delta_{ij}^k| = |\sigma_{ij}^0(x) - \sigma_{ij}^0(a_k)| \le \|\sigma_{ij}^0\|_{\mathcal{H}^\nu} \frac{(b-a)^\nu}{K_\epsilon^\nu}.$$

Choose $\sigma_{ij}^{0k} = \delta_{ij}^k$. Therefore,

$$\left\|\Omega_0^{-1}(\cdot) - \sum_{j=1}^{K_\epsilon} \mathbb{1}_{(a_j, a_{j+1}]}(\cdot)\Omega_{0j}^{-1}\right\|_1 = \int_a^b \left\|\Omega_0^{-1}(x) - \sum_{j=1}^{K_\epsilon} \mathbb{1}_{(a_j, a_{j+1}]}(x)\Omega_{0j}^{-1}\right\| dx$$

$$
\begin{aligned}
&= (b-a)\cdot \mathbb{E}_x \Big\| \Omega_0^{-1}(x) - \sum_{j=1}^{K_\epsilon} \mathbb{1}_{(a_j, a_{j+1}]}(x) \Omega_{0j}^{-1} \Big\| \\
&\le (b-a) \sqrt{\mathbb{E}_x \Big\| \Omega_0^{-1}(x) - \sum_{j=1}^{K_\epsilon} \mathbb{1}_{(a_j, a_{j+1}]}(x) \Omega_{0j}^{-1} \Big\|^2} \\
&= \sqrt{b-a} \sqrt{\int_a^b \Big\| \Omega_0^{-1}(x) - \sum_{j=1}^{K_\epsilon} \mathbb{1}_{(a_j, a_{j+1}]}(x) \Omega_{0j}^{-1} \Big\|^2 dx} \\
&= \sqrt{b-a} \sqrt{\sum_{i=1}^{q} \sum_{j=1}^{q} \int_a^b \Big| \sigma_{ij}^0(x) - \sum_{k=1}^{K_\epsilon} \mathbb{1}_{\Delta_k}(x) \sigma_{ij}^{0k} \Big|^2 dx} \\
&= \sqrt{b-a} \sqrt{\sum_{i=1}^{q} \sum_{j=1}^{q} \int_a^b |\sigma_{ij}^0(x) - \sigma_{ij}(x)| dx} \\
&\le \sqrt{\frac{q^2 H_c (b-a)^{\nu+2}}{K_\epsilon^\nu}} = \epsilon.
\end{aligned}
$$

□

S.2 Proof of Theorem 3.1

Proof. Since all entries of the true covariance matrix $\Omega_0^{-1}(x)$ are in $\mathcal{H}^\nu$, by Lemma 3.1, for a given $\epsilon > 0$, there exists $K > 0$, a set of positive definite matrices $\{\Omega_{0j}^{-1}\}_{j=1}^K$ and $\{a_j\}_{j=1}^K$ where $a_1 = a$, $a_{K+1} = b$, such that

$$
\Big\| \Omega^{-1}(\cdot) - \sum_{j=1}^{K} \mathbb{I}_{(a_j, a_{j+1}]}(\cdot) \Omega_{0j}^{-1} \Big\|_1 < \frac{\epsilon}{3}. \tag{S.2.1}
$$

Since the prior Ω_j^{-1} is absolutely continuous, we have

$$
\Pi\Big\{ \|\Omega_j^{-1} - \Omega_{0j}^{-1}\| < \frac{\epsilon}{3K} \Big\} > 0, \text{ for all } j = 1, \ldots, K.
$$

Hence,

$$
\Pi\Big\{ \sum_{j=1}^{K} \|\Omega_j^{-1} - \Omega_{0j}^{-1}\| < \frac{\epsilon}{3} \Big\} > 0. \tag{S.2.2}
$$

Next, we shall carefully choose neighborhoods B_{μ_j} and $B_{\sigma_j^2}$ so that $\pi_j(x)$ approximates an indicator function. Without loss of generality, assume intervals $(a_j, a_{j+1}]$, $j = 1, \ldots, K$

have equal length $L = (b-a)/K$. We subdivide $(a_j, a_{j+1}]$ into K' equal length intervals $(a_j, a_{j+1}] = \cup_{l=1}^{K'}(a_{jl}, a_{j(l+1)}]$ for $j = 1, \ldots, K$ and the length of $(a_{jl}, a_{j(l+1)}]$ is $\frac{L}{K'}$. Define

$$\pi_{jl}(x) = \frac{\pi_{jl}\frac{1}{\sqrt{\sigma_{jl}^2}}\exp\Big\{-\frac{(x-\mu_{jl})^2}{2\sigma_{jl}^2}\Big\}}{\sum_{t=1}^{K}\sum_{s=1}^{K'}\pi_{ts}\frac{1}{\sqrt{\sigma_{ts}^2}}\exp\Big\{-\frac{(x-\mu_{ts})^2}{2\sigma_{ts}^2}\Big\}},\ j = 1, \ldots, K,\ l = 1, \ldots, K'.$$

Let $\sigma_{jl}^2 \equiv \sigma^2$ and $\pi_{jl} \equiv \pi$ for all j and l, then $\pi_{jl}(x)$ can be simplified to

$$\pi_{jl}(x) = \frac{\exp\Big\{-\frac{(x-\mu_{jl})^2}{2\sigma_{jl}^2}\Big\}}{\sum_{t=1}^{K}\sum_{s=1}^{K'}\exp\Big\{-\frac{(x-\mu_{ts})^2}{2\sigma_{ts}^2}\Big\}},\ j = 1, \ldots, K,\ l = 1, \ldots, K',$$

where μ_{jl} and σ^2 are defined as follows.

$$\mu_{jl} \in \Big[\frac{a_{jl}+a_{j(l+1)}}{2} - \frac{L}{6K'}, \frac{a_{jl}+a_{j(l+1)}}{2} + \frac{L}{6K'}\Big], \qquad \sigma^2 \in \Big[\frac{L^2}{2K'^3}, \frac{L^2}{K'^3}\Big].$$

With absolutely continuous priors on μ_{jl} and σ^2, the prior probabilities of $B_{\mu_{jl}}$ and $B_{\sigma_{jl}^2}$ are non-zero. Let $M = \max\{\|\Omega_{0j}^{-1}\|\}_{j=1}^{K}$. For any $j = 1, \ldots, K$, $l = 1, \ldots, K'$ and the ϵ above, choose

$$K' = \max\left\{\frac{12LM}{5\epsilon}, 30\log\Big(\frac{12LMK}{\epsilon}\Big)\right\} + 1.$$

In the following, we sub-divide $\big\|\pi_{jl}(x) - \mathbb{I}_{(a_{jl}, a_{j(l+1)}]}(x)\big\|_1$ into four mutually exclusive integrals as follows.

$$\begin{aligned}
\Big\|\pi_{jl}(x) - \mathbb{I}_{(a_{jl}, a_{j(l+1)}]}(x)\Big\|_1 &= \int_a^b \Big|\pi_{jl}(x) - \mathbb{I}_{(a_{jl}, a_{j(l+1)}]}(x)\Big|dx \\
&= \int_{x\in\left(a_{jl}, a_{jl}+\frac{L}{5K'}\right]} \Big|\pi_{jl}(x) - 1\Big|dx + \int_{x\in\left(a_{jl}+\frac{L}{5K'}, a_{j(l+1)}-\frac{L}{5K'}\right]} \Big|\pi_{jl}(x) - 1\Big|dx + \\
&\quad \int_{x\in\left(a_{j(l+1)}-\frac{L}{5K'}, a_{j(l+1)}\right]} \Big|\pi_{jl}(x) - 1\Big|dx + \int_{x\notin\left(a_{jl}, a_{j(l+1)}\right]} \Big|\pi_{jl}(x)\Big|dx \\
&:= \mathrm{I}'_{jl} + \mathrm{I}''_{jl} + \mathrm{I}'''_{jl} + \mathrm{I}''''_{jl}.
\end{aligned}$$

First, we have

$$\mathrm{I}'_{jl} \le \frac{L}{5K'}\cdot 1 < \frac{\epsilon}{12\|\Omega_{0j}^{-1}\|}, \qquad \mathrm{I}'''_{jl} \le \frac{L}{5K'}\cdot 1 < \frac{\epsilon}{12\|\Omega_{0j}^{-1}\|}.$$

Moreover,

$$\begin{aligned}
\mathrm{I}''_{jl} &= \int_{x\in\left(a_{jl}+\frac{L}{5K'},a_{j(l+1)}-\frac{L}{5K'}\right]} \left| \frac{1}{1+\sum\sum_{(t,s)\neq(j,l)} \exp\left\{ -\frac{(x-\mu_{ts})^2}{2\sigma^2} + \frac{(x-\mu_{jl})^2}{2\sigma^2}\right\}} - 1 \right| dx \\
&\leq \int_{x\in\left(a_{jl}+\frac{L}{5K'},a_{j(l+1)}-\frac{L}{5K'}\right]} \sum\sum_{(t,s)\neq(j,l)} \exp\left\{ -\frac{(x-\mu_{ts})^2}{2\sigma^2} + \frac{(x-\mu_{jl})^2}{2\sigma^2}\right\} dx \\
&\leq \frac{L}{K'} \cdot KK' \exp\left\{ -\frac{\left(\frac{L}{2K'}-\frac{L}{6K'}+\frac{L}{5K'}\right)^2}{2\frac{L^2}{K'^3}} + \frac{\left(\frac{L}{2K'}-\frac{L}{5K'}+\frac{L}{6K'}\right)^2}{2\frac{L^2}{K'^3}} \right\} \\
&\leq LK \exp\left\{ -\frac{K'}{30}\right\} < \frac{\epsilon}{12\|\Omega_{0j}^{-1}\|},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{I}''''_{jl} &= \int_{x\notin\left(a_{jl},a_{j(l+1)}\right]} \left| \frac{1}{1+\sum\sum_{(t,s)\neq(j,l)} \exp\left\{ -\frac{(x-\mu_{ts})^2}{2\sigma^2} + \frac{(x-\mu_{jl})^2}{2\sigma^2}\right\}} \right| dx \\
&\leq \frac{L}{K'} \exp\left\{ \frac{(x-\mu_{j(l+2)})^2}{2\sigma^2} - \frac{(x-\mu_{jl})^2}{2\sigma^2}\right\}\Bigg|_{x=aj(l+2)} \\
&\leq L \exp\left\{ \frac{\left(\frac{L}{2K'}+\frac{L}{6K'}\right)^2}{2\frac{L^2}{K'^3}} - \frac{\left(\frac{3L}{2K'}-\frac{L}{6K'}\right)^2}{2\frac{L^2}{K'^3}} \right\} \\
&\leq L \exp\left\{ -\frac{2}{3}K'\right\} < \frac{\epsilon}{12\|\Omega_{0j}^{-1}\|}.
\end{aligned}$$

Combining the bounds for I'_{jl}, I''_{jl}, I'''_{jl} and I''''_{jl}, we have

$$\Pi\Big\{ \Big\| \mathbb{1}_{(a_{jl},a_{j(l+1)}]}(\cdot)\Omega_{0j}^{-1} - \pi_{jl}(\cdot)\Omega_{0j}^{-1} \Big\|_1 < \frac{\epsilon}{3} \mid \pi \Big\} > 0, \text{ for all } j=1,\ldots,K, l=1,\ldots,K'.$$

Therefore,

$$\Pi\Big\{ \Big\| \sum_{j=1}^{K}\sum_{l=1}^{K'} \mathbb{1}_{(a_{jl},a_{j(l+1)}]}(\cdot)\Omega_{0j}^{-1} - \sum_{j=1}^{K}\sum_{l=1}^{K'} \pi_{jl}(\cdot)\Omega_{0j}^{-1} \Big\|_1 < \frac{\epsilon}{3} \mid \boldsymbol{\pi} \Big\} > 0. \tag{S.2.3}$$

To finalize, observe by repeated application of triangle inequality,

$$\begin{aligned}
&\Big\| \Omega^{-1}(\cdot) - \sum_{j=1}^{K}\sum_{l=1}^{K'} \pi_{jl}(\cdot)\Omega_{j}^{-1} \Big\|_1 \\
&\leq \Big\| \Omega^{-1}(\cdot) - \sum_{j=1}^{K}\sum_{l=1}^{K'} \mathbb{1}_{(a_{jl},a_{j(l+1)}]}(\cdot)\Omega_{0j}^{-1} \Big\|_1 + \Big\| \sum_{j=1}^{K}\sum_{l=1}^{K'} \mathbb{1}_{(a_{jl},a_{j(l+1)}]}(\cdot)\Omega_{0j}^{-1} - \sum_{j=1}^{K}\sum_{l=1}^{K'} \pi_{jl}(\cdot)\Omega_{0j}^{-1} \Big\|_1 +
\end{aligned}$$

$$\Big\| \sum_{j=1}^{K} \sum_{l=1}^{K'} \pi_{jl}(\cdot)\Omega_{0j}^{-1} - \sum_{j=1}^{K} \sum_{l=1}^{K'} \pi_{jl}(\cdot)\Omega_{j}^{-1} \Big\|_1$$
$$\leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \sum_{j=1}^{K} \sum_{l=1}^{K'} \|\Omega_j^{-1} - \Omega_{0j}^{-1}\| < \epsilon.$$

Apply (S.2.1), (S.2.2) and (S.2.3),

$$\Pi\Big\{ \big\|\Omega^{-1}(\cdot) - \sum_{j=1}^{K} \pi_j(\cdot)\Omega_j^{-1}\big\|_1 < \epsilon \mid \boldsymbol{\pi} \Big\} > 0.$$

By integrating across the prior density of π, we finally conclude the proof. □

S.3 Proof of Theorem 3.2

To prove Theorem 3.2, we simply need to show the prior mass condition holds. To that end, first define the Kullback-Leibler neighborhood of p_{Θ_0} with radius ϵ_n as

$$B_n(p_{\Theta_0}, \epsilon_n) = \Big\{ \Theta \in \mathscr{X} : \int p_{\Theta_0} \log\Big(\frac{p_{\Theta_0}}{p_\Theta}\Big) d\mu \leq \epsilon_n^2, \int p_{\Theta_0} \log^2\Big(\frac{p_{\Theta_0}}{p_\Theta}\Big) d\mu \leq \epsilon_n^2 \Big\},$$

where $\mathscr{X}$ is the parameter space, μ is a dominating measure on the common probability space.

Theorem S.3.1. (Prior Mass Condition). *Suppose Assumption 3.1 and 3.2 hold. Then, for any $C > 0$ and all sufficiently large n,*

$$\Pi\{B_n(p_{\Theta_0}, \epsilon_n)\} \geq \exp(-Cn\epsilon_n^2),$$

where $\epsilon_n = \sqrt{\frac{K(p+q)\log n}{n}}$. Here $\Pi(\cdot)$ is the joint prior probability of all parameters.

Proof. The structure of this proof is similar to the proof of Theorem 3.1 in Norets and Pati (2017), with a few important differences as we shall see below. All constants in the proof are independent of p, q, n, K. Let $\boldsymbol{\mu}_{0j} = (\mu_{0j1}, \ldots, \mu_{0jp})$, $\boldsymbol{\mu}_j = (\mu_{j1}, \ldots, \mu_{jp})$,

$\Sigma_j^x = \mathrm{diag}(\sigma_{j1}^2, \ldots, \sigma_{jp}^2)$. Denote $0 < \lambda_{j1} \leq \ldots \leq \lambda_{jq}$ to be the eigenvalues of Σ_{G_j}. Let $\delta_n = \epsilon_n/\Delta$, where $\Delta = \max\big\{\log(Ke^{-C_6^*(p+q)}), p(e^{-C_1^* q} + e^{-C_2^* p})(\log n + p + q)\big\}$ for Θ in set

$$S_{\Theta_0} = \Big\{(\pi_j, \Sigma_{G_j}, \boldsymbol{\mu}_j, \Sigma_j^x),\ \sum_{j=1}^K |\pi_{0j} - \pi_j| \leq 2\delta_n^2,\ \min_{j=1,\ldots,K} \pi_j \geq \delta_n^4/2, \frac{|\mu_{ji} - \mu_{0ji}|}{\sigma_{0ji}} \leq \delta_n^2,\ \sigma_{ji}^2 \in \Big[\frac{\sigma_{0ji}^2}{1+\delta_n^2}, \sigma_{0ji}^2\Big],$$
$$1 \leq \frac{\lambda_{jk}}{\lambda_{0jk}} \leq 1 + \frac{\delta_n^4}{q},\ i = 1, \ldots, p,\ k = 1, \ldots, q,\ j = 1, \ldots, K\Big\},$$

we have

$$\begin{aligned}
d_H^2\Big\{p(\cdot \mid \Theta_0), p(\cdot \mid \Theta)\Big\} &\leq \Big\| \sum_{j=1}^K \pi_{0j} \cdot \phi_{\Sigma_{G_{0j}}^0} \cdot \phi_{\boldsymbol{\mu}_{0j}, \Sigma_{0j}^x} - \sum_{j=1}^K \pi_j \cdot \phi_{\Sigma_{G_j}} \cdot \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 \\
&\leq \Big\| \sum_{j=1}^K \pi_{0j} \phi_{\Sigma_{G_{0j}}^0} \phi_{\boldsymbol{\mu}_{0j}, \Sigma_{0j}^x} - \sum_{j=1}^K \pi_{0j} \phi_{\Sigma_{G_j}} \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} + \sum_{j=1}^K \pi_{0j} \phi_{\Sigma_{G_j}} \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} - \sum_{j=1}^K \pi_j \phi_{\Sigma_{G_j}} \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 \\
&\leq \sum_{j=1}^K |\pi_{0j} - \pi_j| + \sum_{j=1}^K \pi_{0j} \Big[\Big\| \phi_{\Sigma_{G_{0j}}^0} \phi_{\boldsymbol{\mu}_{0j}, \Sigma_{0j}^x} - \phi_{\Sigma_{G_{0j}}^0} \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 + \Big\| \phi_{\Sigma_{G_{0j}}^0} \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} - \phi_{\Sigma_{G_j}} \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 \Big] \\
&\leq \sum_{j=1}^K |\pi_{0j} - \pi_j| + \sum_{j=1}^K \pi_{0j} (2\pi)^{-\frac{q}{2}} |\Sigma_{G0j}^0|^{-\frac{1}{2}} \Big\| \phi_{\boldsymbol{\mu}_{0j}, \Sigma_{0j}^x} - \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 + \sum_{j=1}^K \pi_{0j} (2\pi)^{-\frac{p}{2}} |\Sigma_j^x|^{-\frac{1}{2}} \Big\| \phi_{\Sigma_{G_{0j}}^0} - \phi_{\Sigma_{G_j}} \Big\|_1 \\
&\leq \sum_{j=1}^K |\pi_{0j} - \pi_j| + \sum_{j=1}^K \pi_{0j} |\Sigma_{G_{0j}}^0|^{-\frac{1}{2}} \Big\| \phi_{\boldsymbol{\mu}_{0j}, \Sigma_{0j}^x} - \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 + \sum_{j=1}^K \pi_{0j} |\Sigma_j^x|^{-\frac{1}{2}} \Big\| \phi_{\Sigma_{G_{0j}}^0} - \phi_{\Sigma_{G_j}} \Big\|_1.
\end{aligned}$$

For $j = 1, \ldots, K$, by Proposition S.3.1,

$$\begin{aligned}
\pi_{0j} |\Sigma_{G_{0j}}^0|^{-\frac{1}{2}} \Big\| \phi_{\boldsymbol{\mu}_{0j}, \Sigma_{0j}^x} - \phi_{\boldsymbol{\mu}_j, \Sigma_j^x} \Big\|_1 &\leq \pi_{0j} \Big(\prod_{i=1}^q \lambda_{ji}^0 \Big)^{-\frac{1}{2}} \Big\{ \sum_{i=1}^p \Big(\frac{\sigma_{ji}^2}{\sigma_{0ji}^2} - 1 - \log \frac{\sigma_{ji}^2}{\sigma_{0ji}^2} \Big) + \sum_{i=1}^p \frac{(\mu_{0ji} - \mu_{ji})^2}{\sigma_{0ji}^2} \Big\}^{\frac{1}{2}} \\
&\leq \pi_{0j} \Big(\prod_{i=1}^q \lambda_{ji}^0 \Big)^{-\frac{1}{2}} \sum_{i=1}^p \Big(\frac{|\mu_{0ji} - \mu_{ji}|}{\sigma_{0ji}} + \Big| \frac{\sigma_{ji}^2}{\sigma_{0ji}^2} - 1 - \log \frac{\sigma_{ji}^2}{\sigma_{0ji}^2} \Big|^{\frac{1}{2}} \Big) \\
&\leq \pi_{0j} \Big(\prod_{i=1}^q \lambda_{ji}^0 \Big)^{-\frac{1}{2}} \sum_{i=1}^p \Big(\frac{|\mu_{0ji} - \mu_{ji}|}{\sigma_{0ji}} + C_1 \Big| \frac{\sigma_{ji}^2}{\sigma_{0ji}^2} - 1 \Big| \Big) \\
&\leq \pi_{0j} (\lambda_m^0)^{-\frac{q}{2}} \cdot p(1 + C_1) \delta_n^2,
\end{aligned}$$

where the second to the last step follows $|\log x - x + 1| \leq C_1 |x - 1|^2$ for x in a neighborhood of 1, where $C_1 > 0$ is a constant. Next, by following the proof from Shen et al. (2013), we show that

$$\pi_{0j} |\Sigma_j^x|^{-\frac{1}{2}} \Big\| \phi_{\Sigma_{G_{0j}}^0} - \phi_{\Sigma_{G_j}} \Big\|_1 \leq \pi_{0j} \Big(\prod_{i=1}^p \sigma_{ji}^2 \Big)^{-\frac{1}{2}} \Big(\Big\| \phi_{\Sigma_{G_{0j}}^0} - \phi_{\Sigma_j'} \Big\|_1 + \Big\| \phi_{\Sigma_j'} - \phi_{\Sigma_{G_j}} \Big\|_1 \Big)$$

$$\leq \pi_{0j}\Big(\frac{\sigma_m^2}{2}\Big)^{-\frac{p}{2}}\left(\Big\|\phi_{\Sigma^0_{G_{0j}}}-\phi_{\Sigma'_j}\Big\|_1+\Big\|\phi_{\Sigma'_j}-\phi_{\Sigma_{G_j}}\Big\|_1\right)\leq 2\pi_{0j}\Big(\frac{\sigma_m^2}{2}\Big)^{-\frac{p}{2}}\delta_n^2,$$

where Σ'_j is a positive definite matrix constructed as follows.

Let $\|\cdot\|_2$ be the matrix spectral norm and $\|\cdot\|_{\max}$ be the maximum of the absolute value of the matrix elements. Assume $\Sigma^0_{G_{0j}}=P_{0j}\Lambda^*_{0j}P_{0j}^T$ and find the orthogonal matrix P_j such that $\Sigma_{G_j}=P_j\Lambda^*_jP_j^T$ and $\|P_{0j}-P_j\|_2\leq\frac{\delta_n^4}{3q\lambda^0_M/\lambda^0_m}$, where $P_{0j}P_{0j}^T=I$ and $P_jP_j^T=I$. Define $\Sigma'_j=P_j\Lambda^*_{0j}P_j^T$ and $Q_j=P_{0j}^TP_j$.

$$\begin{aligned}
&\Big\|\phi_{\Sigma_{G_j}}-\phi_{\Sigma'_j}\Big\|_1\leq\big\{\mathrm{tr}(\Sigma'^{-1}_j\Sigma_{G_j})-\log|\Sigma'^{-1}_j\Sigma_{G_j}|-q\big\}^{\frac{1}{2}}\\
&=\left(\sum_{i=1}^q\frac{\lambda_{ji}}{\lambda_{0ji}}-\log\prod_{j=1}^q\frac{\lambda_{ji}}{\lambda_{0ji}}-q\right)^{\frac{1}{2}}\\
&\leq(q+\delta_n^4-0-q)^{\frac{1}{2}}=\delta_n^2,\\
&\Big\|\phi_{\Sigma'_j}-\phi_{\Sigma^0_{G_0}}\Big\|_1\leq\big\{\mathrm{tr}({\Sigma^0_{G_0}}^{-1}\Sigma'_j)-\log|{\Sigma^0_{G_0}}^{-1}\Sigma'_j|-q\big\}^{\frac{1}{2}}\\
&=\Big\{\mathrm{tr}(P_{0j}{\Lambda^*_{0j}}^{-1}P_{0j}^TP_j\Lambda^*_{0j}P_j^T)-q\Big\}^{\frac{1}{2}}\\
&=\big\{\mathrm{tr}(Q_j\Lambda^*_{0j}Q_j^T{\Lambda^*_{0j}}^{-1}-I)\big\}^{\frac{1}{2}}.
\end{aligned}$$

Let $Q_j=I+B_j$, $\|B_j\|_{\max}\leq\|B_j\|_2=\|P_{0j}^TP_j-I\|_2=\|P_{0j}-P_j\|_2\leq\frac{\delta_n^4}{3q\lambda^0_M/\lambda^0_m}$.

$$\mathrm{tr}(Q_j\Lambda^*_{0j}Q_j{\Lambda^*_{0j}}^{-1}-I)=\mathrm{tr}(B_j+\Lambda^*_{0j}B_j^T{\Lambda^*_{0j}}^{-1}+B_j\Lambda^*_{0j}B_j^T{\Lambda^*_{0j}}^{-1})\leq 3q\|B_j\|_{\max}\frac{\lambda^0_M}{\lambda^0_m}\leq\delta_n^4.$$

Therefore,

$$d_H^2\Big\{p(\cdot\mid\Theta_0),p(\cdot\mid\Theta)\Big\}\leq\left\{2+p(1+C_1)(\lambda^0_m)^{-\frac{q}{2}}+2\Big(\frac{\sigma_m^2}{2}\Big)^{-\frac{p}{2}}\right\}\delta_n^2\leq p(e^{-C_1^*q}+e^{-C_2^*p})\delta_n^2:=C_2\delta_n^2.$$

For any $\Theta\in S_{\Theta_0}$, consider a lower bound on the ratio $p_\Theta/p_{\Theta_0}=p(\mathrm{y},\mathrm{x}\mid\Theta)/p(\mathrm{y},\mathrm{x}\mid\Theta_0)$. Let $a_n=2A_1\sqrt{\log n}+A_2$, where $A_1>0$ and $A_2>0$ are two constants depending on Θ_0, then

$$\mathbb{P}_{\Theta_0}(\|(\mathrm{y},\mathrm{x})\|>a_n)=\sum_{j=1}^K\pi_j\mathbb{P}_{\Theta_0}(\|(\mathrm{y},\mathrm{x})\|>a_n\mid z=j)\leq\frac{1}{n^4}\leq\frac{\delta_n^4}{\log^4(\delta_n^5)},$$

where z indicates which cluster (y,x) is in.

For $\|(\mathrm{x},\mathrm{y})\| \le a_n$, where $\|\cdot\|$ is the Euclidean norm, for all sufficiently large n,

$$p(\mathrm{y},\mathrm{x}\mid\Theta) = \sum_{j=1}^{K}\pi_j\cdot(2\pi)^{-\frac{p}{2}}|\Sigma_j^x|^{-\frac{1}{2}}\exp\left\{-\sum_{i=1}^{p}\frac{(x_i-\mu_{ji})^2}{2\sigma_{ji}^2}\right\}\cdot(2\pi)^{-\frac{q}{2}}|\Sigma_{G_j}|^{-\frac{1}{2}}\exp\left\{-\frac{1}{2}\mathrm{y}^T\Sigma_{G_j}^{-1}\mathrm{y}\right\}$$

$$\ge \min_j\pi_j\cdot(2\pi)^{-\frac{p+q}{2}}(\sigma_M^2)^{-\frac{p}{2}}\lambda_M^{-\frac{q}{2}}\left(1+\frac{\epsilon_n}{q}\right)^{-\frac{q}{2}}\cdot\sum_{j=1}^{K}\exp\left\{-\frac{\|\mathrm{x}-\boldsymbol{\mu}_j\|^2}{2\sigma_m^2}\right\}\exp\left\{-\frac{1}{2}\|\Sigma_{G_j}^{-\frac{1}{2}}\mathrm{y}\|^2\right\}$$

$$\ge (\delta_n^4/2)(2\pi)^{-\frac{p+q}{2}}(\sigma_M^2)^{-\frac{p}{2}}\lambda_M^{-\frac{q}{2}}\left(1+\frac{\epsilon_n}{q}\right)^{-\frac{q}{2}}\sum_{j=1}^{K}\exp\left\{-\frac{\sum_{i=1}^{p}(\epsilon_n\sigma_M+|\mu_{0ji}|)^2}{2\sigma_m^2}\right\}\exp\left\{-\frac{1}{2}\left(\frac{\|\mathrm{x}\|^2}{\sigma_m^2}+\lambda_M\|\mathrm{y}\|^2\right)\right\}$$

$$\ge e^{-C_3^*(p+q)}K\delta_n^4\exp(-C_4^*a_n^2) = C_3K\delta_n^4\exp(-C_4a_n^2),$$

and

$$\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)} \ge e^{-C_5^*(p+q)}K\delta_n^4\exp(-C_4^*a_n^2) := \lambda_n,$$

where $C_3, C_4, C_5 > 0$.

For $\|(\mathrm{x},\mathrm{y})\| > a_n$ and all sufficiently large n,

$$\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)} \ge e^{-C_6^*(p+q)}K\delta_n^4\exp(-C_7^*\|(\mathrm{x},\mathrm{y})\|^2) \ge C_7K\delta_n^4\exp(-C_6\|(\mathrm{x},\mathrm{y})\|^2),$$

where $C_6 = \max\{1/(2\sigma_m^2), \lambda_M/2\}$.

Consider all sufficiently large n such that $\lambda_n \le 1/e$. For any $\Theta \in S_{\Theta_0}$,

$$\int\left(\log\frac{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}{p(\mathrm{y},\mathrm{x}\mid\Theta)}\right)^2\mathbb{1}\left(\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}\le\lambda_n\right)p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y}$$

$$= \int\left(\log\frac{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}{p(\mathrm{y},\mathrm{x}\mid\Theta)}\right)^2\mathbb{1}\left(\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}\le\lambda_n, \|(\mathrm{x},\mathrm{y})\|>a_n\right)p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y}$$

$$\le 2C_7^{*2}\int_{\|(\mathrm{x},\mathrm{y})\|>a_n}\|(\mathrm{x},\mathrm{y})\|^4p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y} + 2\log^2(e^{-C_6^*(p+q)}K\delta_n^4)\int_{\|(\mathrm{x},\mathrm{y})\|>a_n}p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y}$$

$$\le \{2C_7^{*2}(\mathbb{E}_0\|(\mathrm{x},\mathrm{y})\|^4)^{\frac{1}{2}} + 2\log^2(e^{-C_6^*(p+q)}K\delta_n^4)\}[\mathbb{P}_{\Theta_0}(\|\mathrm{x},\mathrm{y})\|\ge a_n)]^{\frac{1}{2}}$$

$$\le 2C_7^{*2}(\mathbb{E}_0\|(\mathrm{x},\mathrm{y})\|^4)^{\frac{1}{2}}\frac{\delta_n^2}{\log^2(\delta_n^5)} + 2\log^2(e^{-C_6^*(p+q)}K\delta_n^4)\frac{\delta_n^2}{\log^2(\delta_n^5)} \le C_8^*\log^2(e^{-C_6^*(p+q)}K)\delta_n^2.$$

Also note that

$$\log\frac{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}{p(\mathrm{y},\mathrm{x}\mid\Theta)}\mathbb{1}\left(\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}\le\lambda_n\right)p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y}$$

$$\leq \left(\log \frac{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}{p(\mathrm{y},\mathrm{x}\mid\Theta)}\right)^2 \mathbb{1}\left(\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)} \leq \lambda_n\right) p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y},$$

Thus,

$$\int \log \frac{p(\mathrm{y},\mathrm{x}\mid\Theta_0)}{p(\mathrm{y},\mathrm{x}\mid\Theta)} \mathbb{1}\left(\frac{p(\mathrm{y},\mathrm{x}\mid\Theta)}{p(\mathrm{y},\mathrm{x}\mid\Theta_0)} \leq \lambda_n\right) p(\mathrm{y},\mathrm{x}\mid\Theta_0)\,d\mathrm{x}\,d\mathrm{y} \leq C_8^* \log^2(e^{-C_6^*(p+q)}K)\delta_n^2.$$

By Proposition S.3.2,

$$\begin{aligned}
\mathbb{E}[\log(p_{\Theta_0}/p_\Theta)] &\leq p(e^{-C_1^* q} + e^{-C_2^* p})\delta_n^2\left(1 + 2\log\frac{1}{e^{-C_5^*(p+q)}K\delta_n^4} + 2C_4^* a_n^2\right) + 2C_8^*\log^2(e^{-C_6^*(p+q)}K)\delta_n^2 \\
&\leq C_9^* p(e^{-C_1^* q} + e^{-C_2^* p})(\log n + p + q)\delta_n^2 + 2C_8^*\log^2(e^{-C_6^*(p+q)}K)\delta_n^2 \leq A\epsilon_n^2, \\
\mathbb{E}[\log(p_{\Theta_0}/p_\Theta)]^2 &\leq p(e^{-C_1^* q} + e^{-C_2^* p})\delta_n^2\left\{12 + 2\left(\log\frac{1}{e^{-C_5^*(p+q)}K\delta_n^4} + C_4^* a_n^2\right)^2\right\} + 8C_8^*\log^2(e^{-C_6^*(p+q)}K)\delta_n^2 \\
&\leq C_{10}^* p(e^{-C_1^* q} + e^{-C_2^* p})(\log n + p + q)^2\delta_n^2 + 8C_8^*\log^2(e^{-C_6^*(p+q)}K)\delta_n^2 \leq A\epsilon_n^2.
\end{aligned}$$

Therefore, both $\mathbb{E}[\log(p_{\Theta_0}/p_\Theta)]$ and $\mathbb{E}[\log(p_{\Theta_0}/p_\Theta)]^2$ are bounded by $A\epsilon_n^2$ for some constant A.

Finally, we calculate a lower bound on the prior probability of $\{\Theta \in S_{\Theta_0}\}$. From Lemma 10 of Ghosal et al. (2007), for some constants $C_{10} > 0$ and all sufficiently large n,

$$\Pi\left(\sum_{j=1}^K |\pi_{0j} - \pi_j| \leq 2\delta_n^2, \min_{j=1,\ldots,K}\pi_j \geq \delta_n^4/2,\right) \geq \exp\{-C_{10}K\log(1/\delta_n)\}.$$

Let m_{ji} be the lower bound of the prior density of μ_{ji} in the interval of $[\mu_{0ji} - \sigma_{0ji}\delta_n^2, \mu_{0ji} + \sigma_{0ji}\delta_n^2]$, $j = 1,\ldots,K, i = 1,\ldots,p$, then

$$\Pi\left(\frac{|\mu_{ji} - \mu_{0ji}|}{\sigma_{0ji}} \leq \delta_n^2,\ j = 1,\ldots,K, i = 1,\ldots,p\right) \geq \prod_{j=1}^K\prod_{i=1}^p m_{ji}2\sigma_{0ji}\delta_n^2 \geq C_{12}\exp\{-C_{11}Kp\log(1/\delta_n)\}.$$

Let m'_{ji} be the lower bound of the prior density of σ_{ji} in the interval of $\left[\frac{\sigma_{0ji}^2}{1+\delta_n^2}, \sigma_{0ji}^2\right]$, $j = 1,\ldots,K, i = 1,\ldots,p$, then

$$\Pi\left(\sigma_{ji}^2 \in \left[\frac{\sigma_{0ji}^2}{1+\delta_n^2}, \sigma_{0ji}^2\right],\ j = 1,\ldots,K, i = 1,\ldots,p\right) \geq \prod_{j=1}^K\prod_{i=1}^p m'_{ji}\delta_n^2\sigma_{oji}^2 \geq C_{14}\exp\{-C_{13}Kp\log(1/\delta_n)\}.$$

Let m''_j be the lower bound of the joint prior density $(\lambda_{j1},\ldots,\lambda_{jq})$ in the region of $[\lambda_{0j1}, \lambda_{0ji}(1 + \delta_n^4/q)] \times \cdots \times [\lambda_{0jq}, \lambda_{0jq}(1 + \delta_n^4/q)]$, then

$$\Pi\left(1 \leq \frac{\lambda_{jk}}{\lambda_{0jk}} \leq 1 + \frac{\delta_n^4}{q},\ j = 1,\ldots,K, k = 1,\ldots,q\right) \geq \prod_{j=1}^K m''_j\prod_{k=1}^q \lambda_{0jk}\delta_n^4/q \geq C_{16}\exp\{-C_{15}Kq\log(1/\delta_n)\}.$$

Therefore, for sufficiently large n, we have

$$\Pi(B_n(p_{\Theta_0}, \epsilon_n)) \geq \Pi(\Theta \in S_{\Theta_0}) \cdot \Pi(G_j, j = 1, \ldots, K) \geq C_{18} \exp\{-C_{17} K(p+q) \log(1/\delta_n)\} \geq \exp(-Cn\epsilon_n^2).$$

□

The conclusion of Theorem 3.2 follows directly from Theorem 3.1 of Bhattacharya et al. (2019).

Proposition S.3.1. (Total variation distance between two normal densities (Devroye et al., 2018)).

$$TV(N(\mu_1, \Sigma_1), N(\mu_2, \Sigma_2)) \leq \frac{1}{2}\sqrt{tr(\Sigma_1^{-1}\Sigma_2 - I) + (\mu_1 - \mu_2)^T \Sigma_1^{-1}(\mu_1 - \mu_2) - \log|\Sigma_1^{-1}\Sigma_2|}.$$

Proposition S.3.2. (Upper bounding KL distance with Hellinger distance (Shen et al., 2013)). *There is a* $\lambda_0 \in (0,1)$ *such that for any* $\lambda \in (0, \lambda_0)$ *and any two probability measures* P *and* Q *with respective densities* p *and* q*,*

$$P \log \frac{p}{q} \leq d_H^2(p,q)\left(1 + 2\log\frac{1}{\lambda}\right) + 2P\left\{\left(\log\frac{p}{q}\right)\mathbb{1}\left(\frac{q}{p} \leq \lambda\right)\right\},$$
$$P\left(\log\frac{p}{q}\right)^2 \leq d_H^2(p,q)\left\{12 + 2\left(\log\frac{1}{\lambda}\right)^2\right\} + 8P\left\{\left(\log\frac{p}{q}\right)^2\mathbb{1}\left(\frac{q}{p} \leq \lambda\right)\right\},$$

where $d_H(\cdot,\cdot)$ *is the Hellinger distance.*

S.4 Posterior Inference

In this section, we develop a blocked Gibbs sampler for the PxG model. Local graph updating procedures are introduced for both approaches in §3.2. We detail the procedures for posterior summary and graph prediction. We also provide the criterion for model comparison between the PxG model and its nested models.

S.4.1 Blocked Gibbs sampler

As stated in §3.1, with the current choice of cohesion function $c(\cdot)$, the DP mixture model is a special case of the PxG model.Therefore, we adopt the blocked Gibbs sampler (Ishwaran and James, 2001) for the PxG model and exploit the (partially) parallel computing for efficient posterior inference. According to Sethuraman (1994), sampling from a DP prior can be constructed using "stick-breaking" weights $V_j \sim \text{Beta}(1, \alpha)$, $j = 1, \ldots, K-1$ and $V_K = 1$ where K is the maximum number of clusters allowed. The weights are then given by $\pi_1 = V_1$ and $\pi_j = V_j \prod_{l=1}^{j-1}(1 - V_l)$, $j = 2, \ldots, K$. The proposed blocked Gibbs sampler consists of five steps at each iteration.

1. Update the posterior stick-breaking weights V_j with

$$V_j \sim \text{Beta}\left(1 + n_j, \alpha + \sum_{l=j+1}^{K} n_l\right), \quad j = 1, \ldots, K-1, \text{ and } V_K = 1.$$

2. Update posterior cluster probabilities π_j with

$$\pi_j = V_j \prod_{l=1}^{j-1}(1 - V_l), j = 2, \ldots, K, \text{ and } \pi_1 = V_1.$$

3. Update cluster indicator z_i for $i = 1, \ldots, n$. Let p_{ij} be the full conditional probability of assigning the ith sample to cluster j. Then we draw z_i from $\{1, \ldots, K\}$ with probability $\{p_{i1}, \ldots, p_{iK}\}$, where

$$p_{ij} = P(z_i = j \mid \cdot) \propto \pi_j L(\mathrm{y}_i \mid \Omega_j, G_j) p(\mathrm{x}_i \mid \boldsymbol{\mu_j}, \sigma_j^2),$$

and $p(\mathrm{x}_i \mid \boldsymbol{\mu}_j, \sigma_j^2)$ is the auxiliary probabilistic model in (3.1). Note that this step can be executed in parallel because of the conditional independencies of z_i's given the cluster probabilities π_j's and the cluster-specific parameters.

4. Update covariate-related parameters $\boldsymbol{\mu}_j$ and σ_j^2, $j = 1, \ldots, K$, from

$$\boldsymbol{\mu}_j \mid \sigma_j^2 \sim N_p(\boldsymbol{\mu}_j^*, \sigma_j^{*2} I_p),$$

$$\sigma_j^2 \sim \text{IG}(b_{1j}^*, b_{2j}^*),$$

where

$$\boldsymbol{\mu}_j^* = \frac{\sigma_0^2}{n_j\sigma_0^2+1}\Big(\sum_{i\in S_j} \mathrm{x}_i + \frac{\boldsymbol{\mu}_0}{\sigma_0^2}\Big), \quad \sigma_0^{*2} = \frac{\sigma_0^2\sigma_j^2}{n_j\sigma_0^2+1},$$

$$b_{1j}^* = \frac{n_j p}{2} + b_1, \quad b_{2j}^* = b_2 + \frac{1}{2}\Big\{\sum_{i\in S_j} \mathrm{x}_i^T\mathrm{x}_i + \frac{\boldsymbol{\mu}_0^T\boldsymbol{\mu}_0}{\sigma_0^2} - \Big(\sum_{i\in S_j}\mathrm{x}_i + \frac{\boldsymbol{\mu}_0}{\sigma_0^2}\Big)^T \boldsymbol{\mu}_j^*\Big\}.$$

When a cluster is empty, draw $\boldsymbol{\mu}_j$ and σ_j^2 from their priors in (3.1).

5. Update cluster-specific parameters Ω_j and G_j for $j = 1, \ldots, K$. This local update depends on the choice of likelihood functions, see the next two subsections for detailed procedures.

S.4.1.1 Local update for the Gaussian-G-Wishart approach

Given samples in the jth cluster, its cluster-specific graphs and precision matrices are updated iteratively for all $1 \le s < t \le q$ in the following two steps.

- Draw the edge inclusion indicator g_{st}^j from its full conditional distribution,

$$P(g_{st}^j = a \mid \cdot) \propto p(\Omega_j \mid G_j^{st,a}) p(G_j^{st,a}) \quad \text{for} \quad a = 0, 1,$$

  where $G_j^{st,a}$ is G_j with $g_{st}^j = a$.

- Draw the precision matrix Ω_j from its full conditional distribution,

$$\Omega_j \sim \text{G-Wishart}_{G_j}(b + n_j, D + \mathrm{Y}_j^{*T}\mathrm{Y}_j^*).$$

S.4.1.2 Local update for the pseudo-likelihood approach

This procedure is carried out independently (in parallel) for q linear regressions. Given the jth cluster, for the sth linear regression, the update contains the following three steps.

- Draw the edge inclusion indicator g_{st}^j from its full conditional distribution,

$$P(g_{st}^j = a \mid \cdot) \propto N(\beta_{st}^j \mid 0, \eta_a \tau_s^j) p(G_j^{st,a}) \quad \text{for} \quad a = 0, 1 \quad \text{and} \quad t \in [q] \backslash s.$$

- Draw the residual variance $\tau_s^j \mid \boldsymbol{\beta}_s^j \sim \text{IG}(a_{1js}, a_{2js})$, where

$$a_{1js} = a_1 + \frac{n_j}{2} + \frac{q-1}{2},$$
$$a_{2js} = a_2 + \frac{1}{2}(\mathrm{Y}_{j,s}^* - \mathrm{Y}_{j,-s}^* \boldsymbol{\beta}_s^j)^T (\mathrm{Y}_{j,s}^* - \mathrm{Y}_{j,-s}^* \boldsymbol{\beta}_s^j) + \frac{1}{2}(\boldsymbol{\beta}_s^j)^T A_{-s}^j \boldsymbol{\beta}_s^j,$$

  and $\mathrm{Y}_{j,s}^*$ is the sth column of Y_j^*; $\mathrm{Y}_{j,-s}^*$ is Y_j^* without its sth column; A_{-s}^j is a diagonal matrix with diagonal entries $1/\eta_{g_{st}^j}$, $t \in [q] \backslash s$.

- Draw the regression coefficient $\boldsymbol{\beta}_s^j \mid \tau_s^j, A_{-s}^j \sim N(\boldsymbol{m}_s^j, \boldsymbol{v}_s^j)$, where

$$\boldsymbol{m}_s^j = \left\{ (\mathrm{Y}_{j,-s}^*)^T \mathrm{Y}_{j,-s}^* + A_{-s}^j \right\}^{-1} (\mathrm{Y}_{j,-s}^*)^T \mathrm{Y}_{j,s}^*,$$
$$\boldsymbol{v}_s^j = \tau_s^j \left\{ (\mathrm{Y}_{j,-s}^*)^T \mathrm{Y}_{j,-s}^* + A_{-s}^j \right\}^{-1}.$$

S.4.1.3 Posterior graph estimation with pseudo-likelihood approach

For efficient computation, we do not constrain $g_{st}^j = g_{ts}^j$ during the course of the Gibbs sampler for the pseudo-likelihood approach. As a result, g_{st}^j and g_{ts}^j are not necessarily the same. To restore the symmetry, we use the common approach that takes either the union or the intersection of g_{st}^j and g_{ts}^j, i.e., $g_{st}^j = g_{ts}^j := g_{st}^j \vee g_{ts}^j$ or $g_{st}^j = g_{ts}^j := g_{st}^j \wedge g_{ts}^j$.

S.4.2 Posterior summary and prediction

The proposed model allows two ways of posterior inference. A clustering result can be obtained directly from the posterior samples of clustering labels. We follow the procedure in Dahl (2006) which minimizes a squared loss function (Binder, 1978). The cluster-specific graphs are then inferred conditionally on the estimated partition.

Alternatively, one can also compute the graph estimate for each observation, i.e., $\widehat{G}(\mathrm{x}_i)$ by the proposed partition averaging technique in §3.3 which is approximated by posterior samples. Specifically, let $\{\widehat{g}_{st}^{\mathrm{x}_i}\}_{s,t=1}^{q}$ denote the edge inclusion indicators of $\widehat{G}(\mathrm{x}_i)$ and let $\{\widehat{p}_{st}^{\mathrm{x}_i}\}_{s,t=1}^{q}$ denote the edge inclusion probabilities where $\widehat{p}_{st}^{\mathrm{x}_i} = P(\widehat{g}_{st}^{\mathrm{x}_i} = 1)$. Then edge inclusion probability is computed by averaging the posterior samples,

$$\widehat{p}_{st}^{\mathrm{x}_i} = \frac{1}{N}\sum_{r=1}^{N}(g_{st}^{z_i^r})^r,$$

where N is the number of posterior samples, z_i^r is the rth posterior sample of the cluster indicator z_i and $(g_{st}^{z_i^r})^r$ is the rth posterior sample of the edge inclusion indicator in cluster z_i^r. The posterior point estimate of $\widehat{G}(\mathrm{x}_i)$ is obtained by $\widehat{g}_{st}^{\mathrm{x}_i} = I(\widehat{p}_{st}^{\mathrm{x}_i} > c)$ for some cutoff $c \in (0,1)$. For simplicity, we use $c = 0.5$ (i.e., the median probability model, Barbieri et al. (2004)) but if desired, c can be also chosen by controlling the posterior expected false discovery rate (Müller et al., 2006). When the data are generated with a definitive clustering structure, such continuous estimates of graphs reflect the estimation uncertainties due to sampling noise. Likewise, the precision matrix function $\widehat{\Omega}(\mathrm{x}_i)$ is estimated by $\widehat{\Omega}_{st}(\mathrm{x}_i) = \frac{1}{N}\sum_{r=1}^{N}(\Omega_{st}^{z_i^r})^r$.

For a new covariate $\mathrm{x}_{\mathrm{new}}$, we can predict its graph $\widehat{G}(\mathrm{x}_{\mathrm{new}})$ and precision matrix $\widehat{\Omega}(\mathrm{x}_{\mathrm{new}})$ by approximating the posterior predictive distribution in (3.3),

$$\widehat{p}_{st}^{\mathrm{x}_{\mathrm{new}}} = \frac{1}{N}\sum_{r=1}^{N}(g_{st}^{z_{\mathrm{new}}^r})^r, \qquad \widehat{g}_{st}^{\mathrm{x}_{\mathrm{new}}} = I(\widehat{p}_{st}^{\mathrm{x}_{\mathrm{new}}} > c), \qquad \widehat{\Omega}_{st}(\mathrm{x}_{\mathrm{new}}) = \frac{1}{N}\sum_{r=1}^{N}(\Omega_{st}^{z_{\mathrm{new}}^r})^r.$$

S.4.3 Model comparison

In order to demonstrate the necessity of incorporating covariates into GGMs with heterogeneous data, we compare the proposed PxG model with two simpler models: (1) partition-based graphical model without covariates, termed *graph-only model*; and (2) random partition model of covariates without the primary variables y, termed *covariate-only*

model. Since they are both nested within the PxG model, we compare them using the deviance information criterion (DIC) (Spiegelhalter et al., 2002; Gelman et al., 2013). DIC accounts for both model fitting and model complexity, and the model with the smallest DIC value is deemed to be the optimum. See the next section for the formulas of all three DICs.

S.4.4 DIC for model comparison

In this section, we give the DIC formula for the PxG model and its two nested model used in model comparison.

DIC for the PxG model (a.k.a. full model)

$$\mathrm{DIC}_{\mathrm{full}}(\hat{\boldsymbol{\rho}}, \hat{G}_1, \ldots, \hat{G}_K) = -2\sum_{j=1}^{\hat{K}} \big(\log p(\mathrm{Y}_j^* \mid \hat{G}_j) + \log g(\mathrm{X}_j^*)\big) + \mathrm{var}\Big(\sum_{j=1}^{\hat{K}} \big(\log p(\mathrm{Y}_j^* \mid \hat{G}_j) + \log g(\mathrm{X}_j^*)\big)\Big),$$

where

$$\mathrm{var}\Big(\sum_{j=1}^{\hat{K}} \big(\log p(\mathrm{Y}_j^* \mid \hat{G}_j) + \log g(\mathrm{X}_j^*)\big)\Big) = \frac{1}{B-1}\sum_{l=1}^{B}\left(\sum_{j=1}^{\hat{K}_l} \big(\log p(\mathrm{Y}_j^* \mid \hat{G}_j^l) + \log g(\mathrm{X}_j^*)\big) - U_0\right)^2,$$

$$U_0 = \frac{1}{B}\sum_{l=1}^{B}\sum_{j=1}^{\hat{K}_l} \big(\log p(\mathrm{Y}_j^* \mid \hat{G}_j^l) + \log g(\mathrm{X}_j^*)\big).$$

B is the number of MCMC samples; $\hat{K}_l$ is the number of clusters in the lth MCMC sample; $\hat{G}_j^l$ is the posterior graph sample of the jth cluster in the lth MCMC sample; $\hat{G}_j$ is the final graph estimate for the jth cluster given cluster allocation $\hat{\boldsymbol{\rho}}$.

DIC for the graph only model (without considering covariates)

$$\mathrm{DIC}_G(\hat{\boldsymbol{\rho}}, \hat{G}_1, \ldots, \hat{G}_K) = -2\sum_{j=1}^{\hat{K}} \log p(\mathrm{Y}_j^* \mid \hat{G}_j) + \mathrm{var}\Big(\sum_{j=1}^{\hat{K}} \log p(\mathrm{Y}_j^* \mid \hat{G}_j)\Big) - 2\log g(\mathrm{X}),$$

where

$$\mathrm{var}\Big(\sum_{j=1}^{\hat{K}} \log p(\mathrm{Y}_j^* \mid \hat{G}_j)\Big) = \frac{1}{B-1}\sum_{l=1}^{B}\left(\sum_{j=1}^{\hat{K}_l} \log p(\mathrm{Y}_j^* \mid \hat{G}_j^l) - U_1\right)^2,$$

$$U_1 = \frac{1}{B}\sum_{l=1}^{B}\sum_{j=1}^{\hat{K}_l}\log p(\mathrm{Y}_j^* \mid \hat{G}_j^l)\big).$$

DIC for the covariate only model (only clustering covariates)

$$\mathrm{DIC}_{\mathrm{cov}}(\hat{\boldsymbol{\rho}}, \hat{G}) = -2\sum_{j=1}^{\hat{K}}\log g(\mathrm{X}_j^*) + \mathrm{var}\Big(\sum_{j=1}^{\hat{K}}\log g(\mathrm{X}_j^*)\Big) - 2\log p(\mathrm{Y} \mid \hat{G}) + \mathrm{var}(\log p(\mathrm{Y} \mid \hat{G})),$$

where

$$\mathrm{var}\Big(\sum_{j=1}^{\hat{K}}\log g(\mathrm{X}_j^*)\Big) = \frac{1}{B-1}\sum_{l=1}^{B}\left(\sum_{j=1}^{\hat{K}_l}\log g(\mathrm{X}_j^*) - U_2\right)^2,$$

$$U_2 = \frac{1}{B}\sum_{l=1}^{B}\sum_{j=1}^{\hat{K}_l}\big(\log\log g(\mathrm{X}_j^*)\big).$$

$$\mathrm{var}(\log p(\mathrm{Y} \mid \hat{G})) = \frac{1}{B-1}\sum_{l=1}^{B}\left(\log p(\mathrm{Y} \mid \tilde{G}_l) - U_2'\right)^2,$$

$$U_2' = \frac{1}{B}\sum_{l=1}^{B}\log p(\mathrm{Y} \mid \tilde{G}_l).$$

For the covariate only model, we assume the graph is the same for all samples. Thus, we perform graph estimation with all samples Y and produce the same amount of MCMC samples; $\tilde{G}_l$ is the posterior graph sample in the lth MCMC sample.

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